Guillaume Berger, PhD

Data Scientist | Control & Robotics Engineer
Optimization • Safety • Artificial Intelligence

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Profile

Data scientist and control and robotics engineer with a PhD in mathematical engineering and 8+ years of R&D experience in control, optimization, AI, and robotics. Expertise in optimization-based and learning-based control and safety-critical systems, from algorithm design to software implementation and experimental validation (end-to-end). Experienced in international multidisciplinary teams.

Professional experience

November 2026–Present
Data Scientist
Octave

- Develop machine learning and optimization tools for EMS

2023–October 2026
FNRS Postdoctoral Researcher
UCLouvain (Louvain-la-Neuve, Belgium)

- Developed control and learning methods for safety-critical autonomous systems
- Supervised PhD researchers and taught **stochastic optimal control and reinforcement learning**

2021–2023
BAEF Postdoctoral Researcher
University of Colorado Boulder (CO, USA)

- Researched learning-based control with safety guarantees for robotic and cyber-physical systems
- Taught **optimization** at master's level

2017–2021
FNRS PhD Researcher
UCLouvain (Louvain-la-Neuve, Belgium)

- Researched control, optimization, and verification of cyber-physical and networked systems with international collaborators

Education

2021 **PhD in Mathematical Engineering**, UCLouvain

2017 **M. Engineering: Applied Mathematics**, UCLouvain
Ranked 1/248; summa cum laude, with congratulations of the jury

2015 **B. Engineering: Applied Sciences**, UCLouvain
Summa cum laude

Programming & computing

Matlab, Python, Julia: Excellent
C/C++: Good

Software development

Git, CI/CD: Good
Docker: Basic
Co-developer of **Dionysos**, an open-source software platform for control of complex systems

Proposal writing

Horizon Europe, NSF, FNRS, Walloon Region, etc.

Languages

French: Native
English: Fluent
Dutch: Basic

Misc

Exchange semester at KTH Stockholm, Sweden (Fall 2016)
Hobbies: Tennis, running, badminton, skiing



Scientific outputs (details on next page)

- 30+ scientific publications in **control, robotics, optimization, machine learning, and formal verification**, including major peer-reviewed international journals and conferences.
- 30+ scientific talks in international conferences and universities (including University of Cambridge, University of Oxford, University of Pennsylvania, École Polytechnique de Paris, and KU Leuven)
- HSCC 2020 Best Paper Award for work on networked cyber-physical systems

Selected publications

- L. Brudermüller, G. Berger, J. Jankowski, R. Bhattacharyya, R. Jungers, N. Hawes, **"CC-VPSTO: Chance-Constrained Via-Point-based Stochastic Trajectory Optimisation for Safe and Efficient Online Robot Motion Planning,"** *International Journal of Robotics Research*, 2026.
- A. Reed, G. Berger, S. Sankaranarayanan, C. Heckman, **"Verified Path Following Using Neural Control Lyapunov Functions,"** *Conference on Robot Learning (CoRL)*, 2022.
- G. Berger, M. Narasimhamurthy, K. Watanabe, M. Lahijanian, S. Sankaranarayanan, **"An algorithm for learning switched linear dynamics from data,"** *NeurIPS*, 2022.
- Z. Wang, G. Berger, R. Jungers, **"Data-Driven Control of Unknown Switched Linear Systems Using Scenario Optimization,"** *IEEE Transactions on Automatic Control*, 2024.
- S. Kamali, G. Berger, S. Sankaranarayanan, **"Synthesizing Min-Max Control Barrier Functions for Switched Affine Systems,"** *Automatica*, 2025.

Full publication list: https://scholar.google.be/citations?user=z_q7fV0AAAAJ&hl=fr

Selected presentations

- September 2025, **University of Oxford**: Invited seminar on recent advances in statistical learning
- May 2025, **UC Irvine**: Invited talk on control of cyber-physical systems with guarantees
- May 2024, **CPS-IoT Week**: Contributed talks on formal methods for robotic systems
- June 2023, **University of Pennsylvania**: Invited talk on learning-based control for cyber-physical systems
- December 2022, **Conference on Robot Learning**: Contributed talk on learning-based control for robotic systems with guarantees

Master-level teaching

- Stochastic Optimal Control and Reinforcement Learning, École Polytechnique de Louvain
- Linear and Integer Programming, CU Boulder College of Engineering and Applied Science

Awards and fellowships

- HSCC 2020 Best Paper Award
- Research Highlight in *Communications of the ACM*
- Postdoctoral researcher fellowship (FNRS)
- Postdoctoral research grant (WBI)
- Postdoctoral researcher fellowship (BAEF)
- PhD fellowship (FNRS–FRIA)

Other academic activities

- Reviewer for 10+ top-tier international journals and conferences
- Member of organizing committee of three international conferences
- Member of IEEE Student Branch at UCLouvain

Internships

- GlaxoSmithKline (Wavre, Belgium): Summer 2014, Summer 2015
- OneSpan (Lommel, Belgium): Summer 2016

Thank you for reading this far! November 11, 2026